

Phases of the Moon

Each night over the course of a month, the Moon appears to change its shape a little. It doesn't actually do so; what we see is differing amounts of the sunlit side as the Moon circles the Earth. These shapes are known as phases of the Moon, and eight phases make up a complete cycle.

The Moon's phases

The complete cycle, from New Moon to Crescent, takes $29\frac{1}{2}$ days. As the Moon appears to grow, it is said to be waxing. As the Moon shrinks, it is said to be waning.

Scale

If the Earth and the Moon were the sizes shown here at the two ends of this page, this would be the distance between them. The Earth's gravity has slowed the Moon's rotation over billions of years, and now one side permanently faces the Earth.



We always see the same face of the Moon.

